SMART INDIA HACKATHON 2024

TITLE PAGE

- **Problem Statement ID – 1648**
- **Problem Statement Title – Online Chatbot based ticketing system.**
- **Theme – Travel & Tourism**
- **PS Category – Software**
- **Team ID – 1363**
- **Team Name – Code Monkeys**

PROPOSED SOLUTION

SOLUTION →

Development of a **Multilingual AI Chatbot** based **Ticketing System** specialized for diverse linguistic communities.

- **Language Detection:** Identify and switch to the user's preferred language.
- **User-Friendly Interface**: Simplify the ticket booking process with easy to use interface.
- **Secure Payment Gateway**: Integrates a payment gateway that support multiple payment methods like UPI, Credit/Debit cards, etc.
- **Automated Email Confirmation**: Sends an email with transaction details, tickets and event information in the user's preferred language.
- **No Form Filling Process:** Chatbot will fill booking form for the users in backend.

UNIQUENESS →

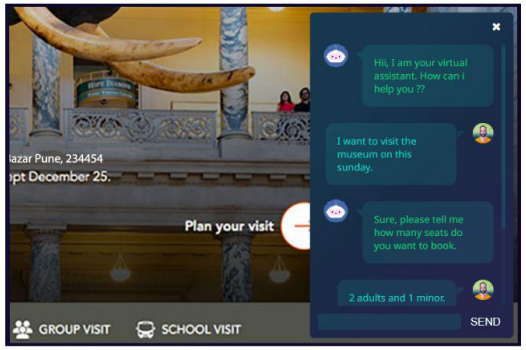
- Use of free and open source AI Model i.e **Meta LLama**.
- Our **Multilingual AI** chatbot simplifies ticket booking with a **form-free**, hassle-free process.
- **Real Time Analytics** to track user behavior and trends, providing data to optimize services and enhance user experience.

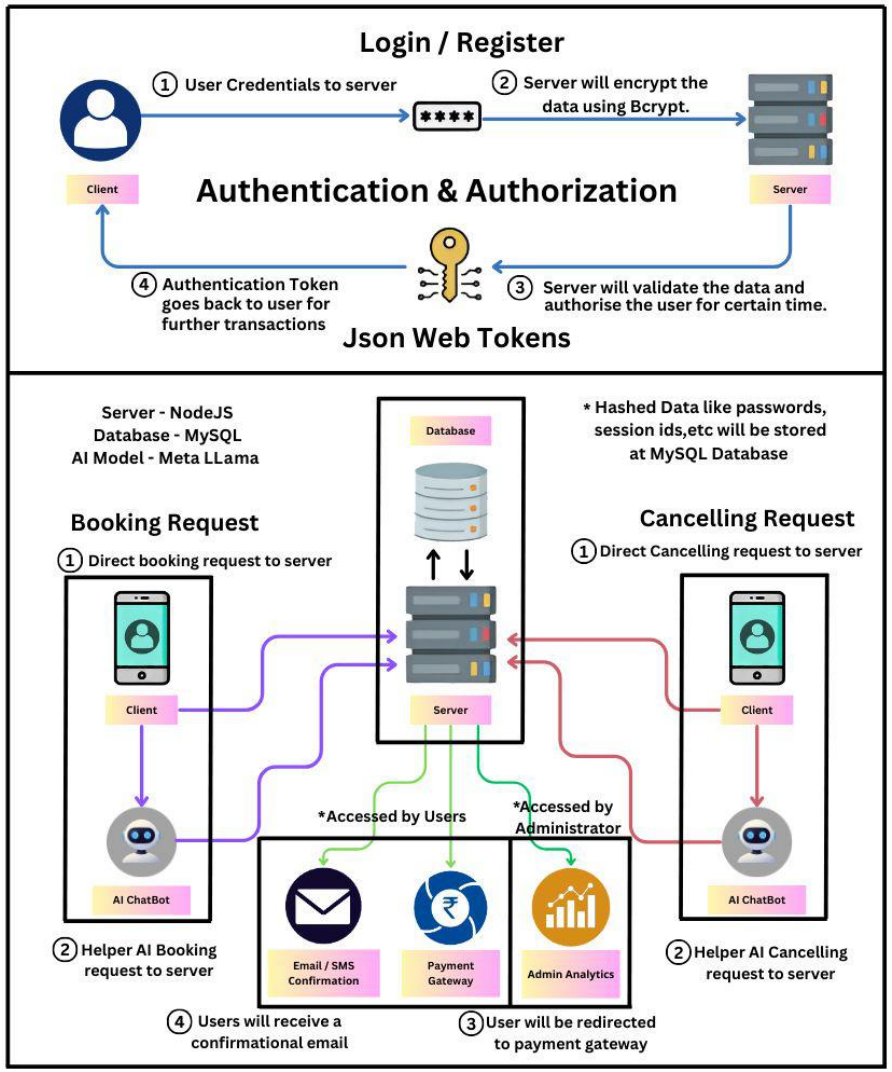
ADDRESSING THE PROBLEM →

Our **multilingual** AI chatbot simplifies museum ticketing, allowing users to buy tickets in their language while boosting efficiency, reducing costs, and enhancing customer experience.

TECHNICAL APPROACH

- **Backend** : Core technologies are **Node JS, Express JS** and **MySQL** database.
- **Frontend** : **React JS** and **Tailwind CSS** will be used to make beautiful User Interface for the website.
- **Cloud** : **Microservices Arch.** Google Cloud Platform.
- **Payment Gateway** : **Razorpay** with UPI, Credit/Debit Card, Netbanking etc.
- **APIs** : Google Translation, Google Analytics, mailgun API, Textlocal API.

Tech Stack			Development in Progress
Front-End	Back-End	Others	
React JS	Node JS	SQL Database	
Tailwind	Express JS	Razorpay UPI Gateway	
Bootstrap	Python for AI	Google Analytics	



FEASIBILITY AND VIABILITY

Analysis of Feasibility →

- The tech stack can scale efficiently by adding resources or more servers to handle more users and data.
- The cost of hosting the web application:-
  - Google Cloud E2 Medium Instance - **₹15,000**/month.
  - Cloud SQL Database - **₹8,000**/month.
  - AI model (META LLAMA) - **₹0** (It will run locally)**.**
  - SMS (TextLocal) - **₹2950/**10000 SMS.
  - Domain name + Mailbox (Hostinger) - **₹750 + ₹1450.**
  - Mailgun - **₹1300/month + ₹150**/1000 emails.

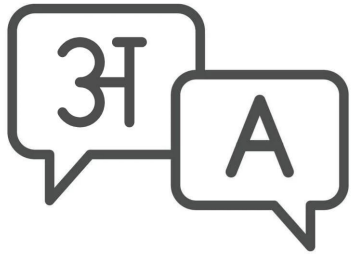
Potential Risks and Challenges →

- **Challenge**: Adding multi-language compatibility to the Large Language Model (AI Chatbot).
- **Mitigation**:Google Translation API can be used instead of training Large Language Model with multiple-languages.

Cost Comparision

ChatGPT	+	AWS	+	APIs	=	₹80,000 / mo
Gemini	+	Google Cloud	+	APIs	=	₹70,000 / mo
LLama	+	Google Cloud	+	APIs	=	₹30,000 / mo
Vertex AI	+	AWS	+	APIs	=	₹65,000 / mo

*More details on research page



IMPACT AND BENEFITS

- Our chatbot system could lead to a **80%** reduction in ticketing errors, such as double bookings or lost records.
- Chatbots can reduce booking time by **75%**, with an average booking time dropping from **10 minutes** to **2.5 minutes**, especially during peak hours.
- Museums implementing chatbot systems can save up to **50%** on **operational costs** by minimizing the need for staff dedicated to ticketing, customer queries, and queue management.
- By accommodating the **90%** of Indian internet users who prefer services in their local language, a **multilingual AI** chatbot makes it simple for people from all over India to buy tickets.
- The chatbot can implement **dynamic pricing** based on factors like time of day or special events, along with **membership programs** for frequent visitors, driving loyalty and offering exclusive benefits and discounts.

Save Operational Cost

Save User's Time

RESEARCH AND REFERENCES

- **LLama research →**
  - https://ai.meta.com/research/publications/the-llama-3-herd-of-models/
- **AI training for chatbot →**
  - https://www.museumnext.com/article/how-are-museums-are-using-chatbots/
  - Dataset of Prototype : https://drive.google.com/file/...
- **Legal Work Research →**
  - https://drive.google.com/file/d/...
- **APIs Cost Comparison →**
  - https://drive.google.com/file/…
- **Payment Gateway Documentation →**
  - https://razorpay.com/docs/payments/payment-gateway
- **Working Prototype Demo →**
  - https://youtu.be/2zsQ9blK_zA